

ASSIGNMENT 2: Backward Propagation Technique

Due: Monday, October 20

Objective: The main objective of this programming assignment is to experience image formation by backward propagation for a simplified 2D case.

Part A: Preliminary test

Consider a single centered point source,

<i>scatter</i>	<i>scatter location</i>
(x_0, y_0)	$(0, 0)$

The *horizontal receiver aperture* is organized in the form of a centered linear receiver array with a span of 60λ (from $x = -30\lambda$ to $x = +30\lambda$). This horizontal linear receiver array is placed at

$$y = -60\lambda.$$

With quarter-wavelength spacing ($\lambda/4$) spacing, there are 241 complex wavefield data samples in total over the 60λ -long linear aperture.

- (i) Perform image reconstruction of the $(60\lambda \times 60\lambda)$ 2D source region. The source region is a square area centered at $(0, 0)$ and bounded by $x = \pm 30\lambda$ and $y = \pm 30\lambda$. For consistency, use quarter-wavelength spacing as the sample spacing in both directions.
- (ii) Obtain and show the 512×512 *FFT spectrum* of the complex image. (Apply the same procedure you used for Assignment 1.)
- (iii) Plot the magnitude distribution of your reconstructed image.

Note that the image reconstruction procedure should be implemented in the form of a one-dimensional convolution along the x direction.

Part B: Multiple point sources

Repeat the exercise with 4 active point sources, located at (x_n, y_n) , $n = 1, 2, \dots 4$.

	<i>scatters</i>	<i>scatter locations</i>
1	(x_1, y_1)	$(0, 15 \lambda)$
2	(x_2, y_2)	$(-12 \lambda, -9 \lambda)$
3	(x_3, y_3)	$(0, -9 \lambda)$
4	(x_4, y_4)	$(+12 \lambda, -9 \lambda)$

Report format:

1. Cover page.
2. Images.
3. Summary: (comments based on your observations)
4. Appendix: (computer code)